

HIERARCHICAL COMPLEXITY IN BIOLOGICAL SYSTEMS

How the Aberconics / GFE Framework Handles Multi-Scale Memory

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Abstract

Biological systems exhibit non-Markovian dynamics across at least eight orders of magnitude in time — from femtosecond bond vibrations to multi-day homeostatic rhythms. Each level of organisation carries its own memory structure, and these levels interact in a nested, bidirectional hierarchy. This document formalises how the Aberconics / GFE framework addresses this challenge through three complementary mechanisms: (i) Sum-of-Exponentials embedding of memory kernels at each level, supported by the complete monotonicity (CM) condition that guarantees physical admissibility; (ii) a hierarchical Memory-Integrated Network (MIN) architecture with an explicit coupling assembly rule governing both bottom-up and top-down interactions; and (iii) a renormalisation flow connecting levels, under which CM and the SOE structure are preserved. The effective dimension D_{eff} is proposed as a conserved-but-renormalised quantity that tracks memory complexity as one coarse-grains across scales. Together, these elements yield a unified, mathematically rigorous, and computationally tractable language for multi-scale biological modelling.

1. The Physical Foundation: Why CM Holds at Every Level

Before introducing the hierarchical architecture, we must establish why the framework's central mathematical property — complete monotonicity of the memory kernel — propagates up the biological hierarchy rather than being an artefact of the molecular-level derivation.

1.1 Complete Monotonicity as a Stability Condition

In the Mori-Zwanzig derivation of the GLE at the molecular level, the memory kernel $K(t)$ is provably CM because it is the autocorrelation function of a bath that is Gaussian, stationary, and in thermal equilibrium. The fluctuation-dissipation theorem ensures $K(t) = (1/kT) F(t)$, where $F(t)$ is the random force autocorrelation — which is positive definite by construction.

At higher levels — cells, tissues, organisms — the bath is neither in equilibrium nor Gaussian. The FDT breaks down. The standard CM proof no longer applies. Yet biological regulatory systems return to homeostasis after perturbation. This stability is the key:

Theorem (CM from Stability):

Let $x(t)$ satisfy a stable linear system near a fixed point with response kernel $K(t)$. Then $K(t)$ is the Laplace transform of a positive Borel measure on $[0, \infty)$ — i.e., $K(t)$ is completely monotone.

Proof sketch: $K(t)$ is the impulse response of a stable, causal, passive system. By Bochner-Bernstein, a function is the Laplace transform of a positive measure iff it is CM. Stability (all poles in left half-plane) combined with passivity (non-negative energy dissipation) implies exactly this.

The biological implication is direct: pathological states — cancer, neurodegeneration, arrhythmia — are precisely states where stability breaks down and CM fails. The CM condition is therefore not just a mathematical convenience. It is the mathematical signature of functional biological organisation.

1.2 SOE Embedding and the Auxiliary Variable System

By Bernstein's representation theorem, every CM kernel admits the decomposition:

$$K(t) = \int_0^\infty e^{-\gamma t} d\mu(\gamma) \approx \sum_\ell w_\ell e^{-\gamma_\ell t}$$

where μ is a positive measure (the spectral measure) and the approximation replaces it with a finite atomic measure at rates $\{\gamma_\ell\}$. This is not merely a fitting trick. Each exponential term corresponds to an auxiliary variable $\chi_\ell(t)$ satisfying:

$$d\chi_\ell/dt = -\gamma_\ell \chi_\ell + u(t)$$

The GLE with memory kernel $K(t)$ is therefore exactly equivalent to a finite-dimensional Markovian system in the extended state space $(u, \chi_1, \chi_2, \dots, \chi_L)$. The dimension of this extension is L , the number of SOE terms. The memory has been made explicit.

The effective memory dimension quantifies how many independent channels are active:

$$D_{\text{eff}} = L \cdot \exp(H_{\text{mem}}) \quad \text{where} \quad H_{\text{mem}} = -\sum_\ell \tilde{w}_\ell \log \tilde{w}_\ell$$

with $\tilde{w}_\ell = w_\ell / \sum w_\ell$. $D_{\text{eff}} = 1$ for a single-exponential (Markovian) kernel. $D_{\text{eff}} = L$ for a uniform spectrum (maximally non-Markovian). Power-law kernels drive $D_{\text{eff}} \rightarrow \infty$ as observation time grows, signalling infinite-dimensional memory.

2. The Duality Theorem: Why Non-Markovian Models Are Complete

The theoretical justification for the entire programme rests on a duality result established in GFE II (Theorem 4.4). This should be understood as the foundational result from which all applications follow.

Theorem 4.4 (GFE II — Nonlinearity-Memory Duality):

(i) Reduction: Any Lipschitz nonlinear Markovian system on a high-dimensional state space, when projected onto a lower-dimensional observable subspace, generates a linear non-Markovian system with a CM memory kernel. The kernel encodes all information about the eliminated degrees of freedom that is observable from the projected subspace.

(ii) Embedding: Conversely, any linear non-Markovian system with CM kernel $K(t)$ can be embedded into a higher-dimensional linear Markovian system via the SOE auxiliary variable construction. The embedding is exact when L is finite and $K(t)$ has a finite atomic spectral measure.

Corollary: The non-Markovian description is not an approximation of the Markovian one — it is exactly equivalent, with the memory kernel carrying all discarded information about hidden degrees of freedom.

The biological interpretation is profound. When we observe a protein conformational change, a synaptic response, or a gene regulatory circuit, we are necessarily working in a projected subspace — we cannot track every atom or every signalling molecule. Theorem 4.4 guarantees that the memory kernel of our observable is the exact reduced description of the full system. Fitting $K(t)$ from data is therefore not a phenomenological act — it is recovering the exact projected dynamics.

This also resolves why cooperative binding, neural firing thresholds, and gene regulatory bistability can all be described within the linear non-Markovian framework. Each nonlinear phenomenon at the full level appears as a specific structure in $K(t)$ at the projected level. The nonlinearity is folded into the memory, not discarded.

3. The Biological Hierarchy and Its Memory Structure

The following table characterises the memory kernel at each level of biological organisation. The D_{eff} values marked with † are theoretical predictions to be validated; those with ‡ are derived from or consistent with existing experimental data.

Level	Timescale	Memory Source	D_{eff}	Observable
Bond / solvent	fs-ps	Velocity autocorr.	1-3 ‡	MD FCF
Residue / backbone	ps-ns	Dihedral transitions	3-10 ‡	NMR R_1/R_2 /NOE
Domain / loop	ns-μs	Conformational exch.	5-15 ‡	NMR Rex, CPMG
Protein complex	μs-ms	Assembly/allostery	8-25 †	smFRET, SPR
Ion channel	ms-s	Open/close/inactivate	2-6 †	Patch clamp
Neuron	ms-100s	Synaptic + intrinsic	10-25 †	Intracellular

Level	Timescale	Memory Source	D_eff	Observable
Neural population	100ms–min	Plasticity, adaptation	25–100 †	LFP, EEG
Cell / signalling	s–min	Feedback loops	15–50 †	Live-cell imaging
Tissue / organ	min–hrs	Mechanical memory	20–80 †	Rheology
Organism	hrs–days	Homeostatic feedback	≤80† per pathway	Physiology

‡ Derived from or consistent with NMR/MD data in the literature. † Theoretical predictions of the renormalisation flow.

4. Hierarchical MIN Architecture with Explicit Coupling

We formalise the hierarchical Memory-Integrated Network as a nested operator structure. Critically, we specify the coupling assembly rule — the mechanism by which the slow (outer) level modulates the fast (inner) level's dynamics, which was the principal gap in previous formulations.

4.1 The Nested Volterra Structure

A hierarchical MIN is defined as a composition of causal integral operators:

$$\mathcal{M}[\mathbf{f}] = \sum_{\ell} \prod_j v_{\ell,j,1} (v_{\ell,j,2} (\dots v_{\ell,j,r} [\mathbf{f}] \dots))$$

where each v is a causal Volterra operator with its own CM kernel $K_v(t)$. The inner layers correspond to fast processes (ion channel fluctuations, fast backbone dynamics); the outer layers correspond to slow processes (synaptic plasticity, gene regulation). SOE embedding converts each v into a finite-dimensional ODE, making the full hierarchy computationally tractable.

4.2 Bottom-Up Coupling: Fast Drives Slow

At level n , the coarse-grained observable X_n evolves under a GLE whose forcing term includes the output of the level below:

$$dX_n/dt = -\nabla V_n(X_n) - \sum_{\ell} w_{\ell}^n \chi_{\ell}^n + F_n(X_{n-1}, t)$$

where F_n is a coupling function of the level-($n-1$) output, and $\{\chi_{\ell}^n\}$ are the level- n auxiliary variables. This is bottom-up coupling: fast processes create the effective forcing and noise at the slow level. It is the standard Mori-Zwanzig coarse-graining, and it preserves CM at level n provided $K_{n-1}(t)$ is CM.

4.3 Top-Down Coupling: Slow Modulates Fast — The Assembly Rule

Top-down modulation is the harder case and the one previously unspecified. The slow variable $X_n(t)$ modulates the memory kernel of the fast level $n-1$. We formalise this through a parameter-dependent kernel:

$$K_{n-1}(t; X_n) = \sum_{\ell} w_{\ell}(X_n) \cdot e^{-\gamma_{\ell}(X_n) t}$$

Both the weights w^ℓ and the rates γ^ℓ at level $n-1$ are functions of the slow variable X_n . This is the biologically correct picture: the slow conformational state of a protein modulates the fast backbone fluctuations; the slow gene regulatory state modulates the fast ion channel kinetics; the slow mechanical state of tissue modulates the fast cellular dynamics.

The CM condition requires that for each fixed value of X_n , the kernel $K_{\{n-1\}}(t; X_n)$ remains CM. This imposes the constraint:

$$w^\ell(X_n) \geq 0 \quad \text{and} \quad \gamma^\ell(X_n) > 0 \quad \text{for all } X_n$$

These are simple inequality constraints on the coupling functions. They can be enforced by parameterising $w^\ell = \exp(\alpha^\ell(X_n))$ and $\gamma^\ell = \exp(\beta^\ell(X_n))$ with unconstrained α, β — or by NNLS at each update step. The result is a CM-preserving top-down coupling that is both biologically meaningful and mathematically consistent.

Coupling Assembly Rule (complete specification):

Level n drives level $n-1$ via forcing: $F_{\{n-1\}}(t) \leftarrow g(X_n(t))$

Level n modulates level $n-1$ kernel: $K_{\{n-1\}}(t) \leftarrow K_{\{n-1\}}(t; X_n(t))$

Level $n-1$ drives level n via output: $F_n(t) = h(X_{\{n-1\}}(t))$

Coupling constraints: $w^\ell(X_n) \geq 0, \gamma^\ell(X_n) > 0$ (CM preserved)

g and h are learned from data using local Hebbian rules (cross-level correlation).

The full hierarchy is a coupled ODE system with total dimension $\sum_n L_n$.

5. Renormalisation Flow Across the Biological Hierarchy

This is the new theoretical contribution that was absent from previous formulations. We establish a precise mathematical relationship between the memory kernel at level n and the kernel at level $n+1$, and show that this relationship constitutes a renormalisation group flow in the space of CM kernels.

5.1 The Coarse-Graining Operation

Moving from level n to level $n+1$ involves integrating out the fast degrees of freedom at level n , retaining only the slow observable X_n . This is a coarse-graining or Mori-Zwanzig projection. The resulting kernel at level $n+1$ is:

$$K_{\{n+1\}}(t) = K_n(t) * K_{\text{bath}}^n(t) + K_{\text{direct}}^n(t)$$

where $*$ denotes temporal convolution, $K_{\text{bath}}^n(t)$ is the bath correlation function at level n , and $K_{\text{direct}}^n(t)$ captures direct memory contributions that survive the coarse-graining. This is the renormalisation map R_n :

$$R_n : K_n \mapsto K_{\{n+1\}} = K_n * K_{\text{bath}}^n + K_{\text{direct}}^n$$

5.2 Preservation of CM Under Coarse-Graining

The key structural result justifying the use of the same framework at every level:

Proposition (CM is preserved under R_n):

If $K_n(t)$ is CM and $K_{\text{bath}}^n(t)$ is CM, then $K_{\{n+1\}}(t) = K_n * K_{\text{bath}}^n$ is CM.

Proof: The convolution of two CM functions is CM. This follows from the fact that the Bernstein class is closed under pointwise products (since $K_n(t) \cdot K_{\text{bath}}^n(t)$ is CM for CM functions), and convolution corresponds to multiplication of Laplace transforms. The product of two completely monotone Laplace transforms is completely monotone. $\square$

Corollary: CM propagates upward through the entire biological hierarchy. Provided each level is individually stable (its bath is CM), the effective kernel at every higher level is also CM.

5.3 Renormalisation of SOE Parameters

Under R_n , the SOE parameters transform in a computable way. If $K_n(t) = \sum_l w_l \exp(-\gamma_l t)$ and $K_{\text{bath}}^n(t) = \sum_m u_m \exp(-\delta_m t)$, then their convolution is:

$$K_{\{n+1\}}(t) = \sum_l \sum_m (w_l u_m / (\gamma_l - \delta_m)) \cdot [e^{-\delta_m t} - e^{-\gamma_l t}] \quad (\gamma_l \neq \delta_m)$$

This is still a sum of exponentials with $L \times M$ terms — but many will have small weights and can be pruned. The effective number of channels at level $n+1$ is therefore at most $L \times M$, but typically much smaller after pruning. The rates at level $n+1$ are the union of $\{-\gamma_l\}$ and $\{-\delta_m\}$ — the slow bath rates dominate, the fast signal rates are suppressed.

This is the renormalisation flow in parameter space: each level inherits the slow channels from the level below and generates new slow channels from the bath correlation. Fast channels are integrated out. The SOE spectrum shifts toward slower timescales as we ascend the hierarchy.

5.4 Renormalisation of D_{eff}

How does D_{eff} transform under R_n ? This is the central diagnostic of the renormalisation flow.

Because $K_{\{n+1\}}$ has at most $L \times M$ channels but typically fewer, and because the weight distribution shifts toward the new slow channels, we can derive bounds:

$$D_{\text{eff}}^{n+1} \leq D_{\text{eff}}^n \cdot D_{\text{eff}}(K_{\text{bath}}^n)$$

with equality when all channels have identical weight (maximum entropy case). In biological systems, the bath correlation is typically dominated by a small number of slow modes — $D_{\text{eff}}(K_{\text{bath}}) \sim 2-5$. So D_{eff} grows sub-multiplicatively with each level, consistent with the observed moderate D_{eff} values at each biological scale.

Critically, D_{eff} at level $n+1$ is not simply D_{eff} at level n plus new channels. It is a renormalised quantity whose entropy reflects the new weight distribution after coarse-graining. Channels that were nearly degenerate at level n may merge into a single effective channel at level $n+1$, reducing D_{eff} . New slow channels from the bath may increase it.

Summary: Renormalisation Flow Properties

1. CM is preserved at every level (Proposition above).
2. SOE structure is preserved at every level (convolution of SOEs is SOE).
3. Fast channels are suppressed; slow channels accumulate upward.
4. D_{eff} grows sub-multiplicatively: $D_{\text{eff}}^{n+1} \leq D_{\text{eff}}^n \cdot D_{\text{eff}}(\text{bath}^n)$.
5. Pathological states (cancer, neurodegeneration) correspond to fixed points or divergences of the flow where D_{eff} grows without bound or collapses to 1.

6. Spectral Units as Common Currency Across Scales

The spectral units derived from $K(t)$ provide a universal vocabulary that allows comparison across levels, systems, and experimental modalities. They are well-defined at every level of the hierarchy and transform predictably under the renormalisation flow.

Spectral Unit	Definition and Biological Meaning
D_{eff}	Effective number of independent memory channels. Quantifies dynamical complexity. Conserved-but-renormalised across levels.
M_{cap} (mean depth)	Mean memory timescale $\Sigma \tilde{w}_l / \gamma_l$. Sets the dominant integration window of the system. Shifts to slower values at higher levels.
M_{scale} (span)	$\log_{10}(\gamma_{\text{max}} / \gamma_{\text{min}})$. Number of decades covered. Grows monotonically up the hierarchy.
H_{mem} (entropy)	Shannon entropy of normalised weights. Maximum when all channels contribute equally. High H_{mem} = rich, distributed memory.
d_s (fractal exponent)	Spectral dimension from FACF power-law tail. $d_s < 1$ signals infinite-dimensional memory approaching a branch cut.

These units enable statements that cross disciplinary boundaries. A condensate made of proteins with $D_{\text{eff}} = 15$ has quantifiably richer memory than one with $D_{\text{eff}} = 3$ — and the renormalisation flow predicts how that cellular-level D_{eff} emerges from the molecular-level kernels of its constituent proteins.

7. Handling Unbounded Complexity: The Infinite-Dimensional Limit

Certain biological phenomena — the rheology of cytoplasm, the richness of neural population dynamics, evolutionary adaptation — exhibit memory that is effectively infinite-dimensional. The framework diagnoses and handles this case explicitly.

A power-law kernel $K(t) = t^{-\alpha}$ with $0 < \alpha < 1$ is CM, but its Laplace transform has a branch cut on the negative real axis rather than discrete poles. The SOE approximation converges, but $D_{\text{eff}} \rightarrow \infty$

as the number of terms L grows with observation time. No finite-dimensional embedding captures the full dynamics; any practical model is necessarily truncated.

The fractal exponent $d_s = 2\alpha$ provides the diagnostic:

$$K(t) \sim t^{-d_s/2} \Rightarrow J(\omega) \sim \omega^{d_s/2-1} \Rightarrow D_{\text{eff}}(T_{\text{obs}}) \sim T_{\text{obs}}^{1-d_s/2}$$

This allows principled truncation: choose L such that the truncation error in $J(\omega)$ at the highest frequency of interest is below a specified tolerance. The error bound is:

$$\varepsilon(L) = \int_{\gamma_L}^{\infty} d\mu(\gamma) / (\gamma^2 + \omega_{\text{max}}^2) \leq \mu([\gamma_L, \infty)) / \omega_{\text{max}}^2$$

This is the renormalisation group fixed point of the flow for fractal systems: the kernel is scale-invariant, and every level of the hierarchy sees the same power-law memory with the same exponent, only stretched in time. Turbulent blood flow, cytoplasmic viscoelasticity, and certain neural dynamics belong to this universality class.

| 8. Blueprint for Multi-Scale Biological Modelling

The following pipeline integrates all elements of the framework into a practical protocol for modelling any biological system exhibiting hierarchical non-Markovian dynamics.

- **Step 1 — Data acquisition.** Gather time-series data at each relevant scale: single-channel currents, NMR relaxation, spike trains, live-cell imaging, tissue rheology. Identify the observable at each level and the relevant timescale range.
- **Step 2 — Kernel extraction.** For each level, extract $K(t)$ via the appropriate method: FACF from MD; R_1/R_2 /NOE inversion (two-stage pipeline) from NMR; impulse response from electrophysiology. Run the CM filter. Fit SOE via joint NNLS with Rex extraction if chemical exchange contributes. Compute spectral units.
- **Step 3 — Renormalisation consistency check.** Verify that D_{eff} grows sub-multiplicatively between levels. Check that M_{scale} increases monotonically. If D_{eff} decreases between levels, this signals either channel merging (physical) or fitting artefact (methodological) — use Prony diagnostics to distinguish.
- **Step 4 — Hierarchical MIN construction.** Couple levels using the assembly rule: bottom-up via forcing $F_n(X_{n-1})$, top-down via parameter modulation $K_{n-1}(t; X_n)$. Enforce CM preservation at each coupling: $w_l(X_n) \geq 0$, $\gamma_l(X_n) > 0$. Learn coupling functions from cross-level correlation data using local Hebbian rules.
- **Step 5 — Simulation.** Integrate the coupled auxiliary variable ODE system. Total dimension = $\sum_n L_n$, typically 20–100 for a 3–4 level model. Run Formulation B dynamics (Aberconics u, m, v decomposition) for each level's observable.
- **Step 6 — Tuning and inverse design.** Tune individual fields (u, m, v) at any level to explore the dynamical neighbourhood of the real system. Specify a target behaviour at level $n+1$ and invert the renormalisation map to determine which level- n kernel parameters

produce it. For sequence-level design, invert the Rouse eigenvalue problem to recover residue composition constraints.

- **Step 7 — Validation.** Compare predictions at each level against held-out experimental data. Use spectral units as the primary validation metric: predicted D_{eff} , M_{cap} , and d_s should match observed relaxation spectra. Mismatches diagnose specific modelling failures: D_{eff} too low $\rightarrow$ insufficient memory channels; M_{scale} too narrow $\rightarrow$ missing slow or fast channels; d_s wrong $\rightarrow$ incorrect fractal structure.

| 9. Conclusions

The Aberconics / GFE framework addresses biological hierarchical complexity not by treating each level independently, but by establishing a single mathematical object — the CM memory kernel and its SOE representation — that is well-defined, physically meaningful, and computationally tractable at every level of biological organisation.

Three results carry the weight of the formal argument. The CM-from-stability theorem justifies applying the framework above the equilibrium molecular level. The nonlinearity-memory duality theorem guarantees that linear non-Markovian models are complete descriptions of projected nonlinear systems. The renormalisation flow shows that CM and the SOE structure propagate upward under coarse-graining, with D_{eff} growing sub-multiplicatively and the spectral weight shifting systematically toward slower timescales.

The result is not a collection of level-specific models connected by ad hoc interfaces. It is a tower of equivalent theories, one at each scale, connected by mathematically derived transformations that preserve the framework's core properties. The spectral units provide the common language in which measurements at any level can be compared and predictions at other levels derived.

The immediate frontier is the experimental validation of the renormalisation flow: measuring D_{eff} at the residue, domain, and complex levels for a single protein system and verifying the predicted sub-multiplicative scaling. The CaM pipeline we have built — extracting $K(t)$ from NMR data at the residue level — is the starting point for that measurement.

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